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EXPANDABLE FLIGHT FOR A CONVEYOR

Abstract

A conveying system employs expandable flights for controlling the spacing of objects conveyed by a conveyor. Each flight includes a body portion comprising a pair of panels separated by an expandable linkage system. The flight is pivotally mounted to a mover of a linear transport system via a mounting portion so that the flight can be moved independently of the conveyor.

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Background/Summary

RELATED APPLICATIONS [0001] The present application claims priority to U.S. Provisional Patent Application No. 63/334,754, entitled “Expandable Flight for a Conveyor”, filed Apr. 26, 2022, the contents of which are herein incorporated by reference.

BACKGROUND OF THE INVENTION

[0002] The present invention relates to the field of power-driven conveyors. More particularly, the invention relates to a system and method for controlling the spacing of conveyed objects.

[0003] It is often desirable to space conveyed objects in a selected manner along the travel direction of a conveyor, such as a conveyor belt. The spacing enables downstream processing, such as sorting by diverting of selected objects onto an exit conveyor. For example, it may be desirable to have only one package on a section of a conveying system, such as a diverter, at a time. If the packages are of different sizes, such as often occurs in the shipping industry, then the spacing of packages should be varied in order to ensure that only one package is on the selected section at a time, while maintaining the smallest possible gap between packages. In addition, non-regular packages, such as envelopes and bags may be difficult for flights used to space the conveyed objects to operate.

SUMMARY OF THE INVENTION

[0004] In one embodiment, a flight for a conveyor comprises an expandable body and a mounting portion for mounting the expandable body relative to the conveyor. The expandable body comprises a first panel, a second panel and a linkage system between the first panel and the second panel for selectively moving the second panel relative to the first panel. The mounting portion is connected to the first panel.

[0005] In another embodiment, a conveyor system for conveying and spacing articles comprises a conveyor having a conveying surface extending longitudinally from a first end to a second end and a plurality of flights extending laterally over the conveying surface. Each flight has an expandable body that can expand longitudinally.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0006] FIG. 1 is an isometric view of a dynamic gapping conveying system according to an embodiment;

[0007] FIG. 2 is an isometric view of the dynamic gapping conveying system of FIG. 1 with parcels gapped and conveyed;

[0008] FIG. 3 is an isometric view of a flight with an expandable body in the dynamic gapping conveying system of FIG. 1;

[0009] FIG. 4 is a detailed view of the body of the flight of FIG. 3 in a collapsed position;

[0010] FIG. 5 is a detailed view of the body of FIG. 4 in a partially expanded position;

[0011] FIG. 6 is a detailed view of the body of FIG. 4 in an expanded position;

[0012] FIG. 7 is an exploded view of the body of FIG. 4;

[0013] FIG. 8 is a top view of the body of FIG. 4;

[0014] FIG. 9 is a detailed view of a hinged connection between two flat links of the body of FIG. 4;

[0015] FIG. 10 is another view of the hinged connection of FIG. 9;

[0016] FIG. 11 is a detailed view of a latch for securing the body of FIG. 4 in an expanded position;

[0017] FIG. 12 is an isometric transparent view of the actuator for the linkage system of the flight of FIG. 3.

DETAILED DESCRIPTION OF THE INVENTION

[0018] A conveying system includes a dynamic, expandable flight to control and vary the spacing of conveyed objects. The present invention will be described below relative to certain illustrative embodiments. Those skilled in the art will appreciate that the present invention may be implemented in a number of different applications and embodiments and is not specifically limited in its application to the particular embodiments depicted. FIGS. 1 and 2 show a dynamic gapping conveying system **100** employing dynamic, expandable flights. The dynamic gapping conveying system includes a conveyor **120** that moves objects from the first end **111** to the second end **112** in a conveying direction **12**. A dynamic gapping system **130** runs concurrent with the conveyor **120** and includes independent flights **160** for spacing objects conveyed by the conveyor. For example, the dynamic gapping system **130** may control and maintain a consistent gap between packages **102** of different sizes to increase throughput. The dynamic gapping system **130** comprises a linear transport system connected to a plurality of flights **160** that extend laterally over the conveyor conveying surface **122**. The illustrative flights **160** are controlled separately from the conveying surface **122**. The illustrative flights **160** are expandable to adjust a gap between conveyed parcels **102**, as shown in FIG. 2. If a smaller gap is desired, a flight **160** may be compressed. If a larger gap is desired, a flight **160** may be expanded to provide a selected gap size.

[0019] The illustrative conveyor **120** comprises as an endless conveyor belt **120** trained around guide devices at each end **111**, **112**, defining a top conveying surface **122** and forming a returnway below the carryway to form a complete circuit. A driver, such as a sprocket driven by a motor, moves the conveyor belt **120** through the circuit to move objects from the first end **111** to the second end **112**. The conveyor carryway extends longitudinally from the first end **111**, which is the receiving end, to the second end **112**, which is the discharge end, and laterally in width from a first side edge to a second side edge.

[0020] The invention is not limited to a conveyor belt for conveying objects, and any suitable means for conveying objects may be used. For example, the conveyor can comprise a low friction surface, rollers or another suitable conveyor.

[0021] The dynamic gapping system **130** includes a plurality of flights **160** extending laterally across the width, or a portion of the width, of the conveying surface **122**, perpendicular to the conveying direction **12**, for guiding conveyed objects. The flights **160** can serve as stops to limit the travel of objects **102**, such as packages, to control their relative spacing on the conveying surface. Alternatively, or in addition, the flights **160** may serve as pushers for pushing the objects in the conveying direction.

[0022] In an illustrative embodiment of the invention, the flights **160** are driven independently from the conveying surface **122** and, in certain embodiments, each other. The flights **160** may be separated by a variable separation distance, which allows the flights **160** to be used to independently vary object spacing. The position of the flights **160** is longitudinally adjustable relative to the conveying surface **122** and-or each flight **160** may travel at a different speed than the conveying surface **122**.

[0023] The illustrative flights **160** can be oriented in an operating position over the carryway. The illustrative flights can then be rotated and compressed to allow for travel around the reversing elements at the end of the carryway and through the conveyor returnway. Then, when a flight reenters the carryway, the flight **160** can rotate and expand to form a block for parcels **102** to set a predefined gap, as shown in FIG. 2.

[0024] In one embodiment, the flights **160** are driven by a linear transport system, comprising a plurality of movers **133** connected to the flights **160** and a motor module, such as a brushless BLDC motor or other type of linear magnetic motor, for propelling the movers, with each mover individually controlled. Each illustrative motor module comprises an endless, obround-shaped rail **132** on a side of the conveyor belt, substantially matching and adjacent the path of the conveyor belt **120**, though the invention is not so limited. Alternatively, the rail **132** could return along a different path, such as above or outside of the carryway. The illustrative rail **132** includes inducers,

such as embedded electromagnetic coils or other elements, that cooperate with elements, such as magnetic plates, in the movers **133** to propel the movers through the circuit formed by the rail **132** at a controlled and variable pace. Suitable linear transport systems are available from Rockwell Automation (the iTRACK® intelligent track system), Beckhoff Automation LLC of Savage, MN, B&R Automation of Eggelsberg, Austria, FESTO Corporation of Germany and other linear transport system providers known in the art.

[0025] Referring to FIG. **3** each illustrative flight **160** comprises an expandable body portion **161** and a mounting portion **162** that connects the body portion **161** to a mover **133** of the linear transport system so that when the mover **133** travels along the rail **132**, the connected body portion **161** moves, too, creating dynamically assignable flights across the conveying surface **122**. In one embodiment, the body portion **161** is pivotally mounted to the mover **133** to allow the body portion **161** to pivot relative to the mover **133** and the conveying surface **122**. The illustrative mounting portion **162** is a mounting bar rotatably mounted in a sheath **164** on the mover **133**.

[0026] The illustrative flight includes follower rollers **166**, **167** at each end for engaging a track in the frame of the conveying system to control the rotational position of the flight. For example, the track and follower rollers may rotate the body **161** relative to the mover **133** to allow the flight to fit into a small space at the end of the carryway.

[0027] As shown in FIGS. **4-8**, the expandable body portion **161** comprises a first panel **171** connected to the mounting portion **162** and a second panel **172** that is movable longitudinally relative to the first panel **171**. Each panel extends laterally from an inner end adjacent the mounting portion **162** to a distal end and from a top edge to a bottom edge that is configured to extend over a conveying surface **122**. Each panel is substantially flat and thin in thickness relative to height and width. A linkage mechanism selectively expands and collapses the body portion by moving the second panel **172** away from and towards the first panel **171**. The linkage mechanism comprises an input link **181** pivotally connected at a first end to a retractable rod **180**. The retractable rod **180** is housed in the mounting portion **162**. The end of the mounting portion forms a channel **168** for exposing the end of the retractable rod. The tip **163** of the mounting portion is fixed to a first end of the first panel **171**.

[0028] The input link **181** is pivotally connected at a second end to the second panel **172**, preferably around the lateral middle of the second panel and at a top edge of the second panel **172**, but the invention is not so limited. In an embodiment, the second panel **172** includes a pin that is received in a hinge knuckle at the second end of the input link **181**.

[0029] A cross link **183** pivotally connects at a first end to a middle of the input link **181** and at a second end to the first panel. The first panel **171** may include a pin that is received in a hinge knuckle at the second end of the cross link **183**. When the rod **180** is retracted within the mounting portion **162**, the rod pulls the second panel **172** towards and adjacent to the first panel **171** so that the body is collapsed, as shown in FIGS. **4** and **8**. To expand the body **161**, the rod **180** can extend, as shown in FIGS. **5** and **6**, pushing the second panel **172** away from the first panel **171**.

[0030] Reinforcing links can stiffen the body in the expanded position, while minimizing the thickness of the body when the panels are in the collapsed position, as shown in FIG. **8**. The illustrative reinforcing links, shown in FIGS. **5-7**, comprise a first set of flat links **191**, **192** pivotally connected to each other and the panels **171**, **172**. The first flat link **191** is pivotally connected at a first end to the first panel **171** below the input link **181**. The first flat link **191** extends to an inner surface of the second panel **172**. The second flat link **192**, shorter than the first flat link **191**, extends below the first flat link **191** and is pivotally connected at a first end to a side edge of the second panel **172**. The second end of the second flat link **192** is pivotally connected to the first flat link **191** by a hinge knuckle **198** that receives a pin **199** on the first flat link **191**, as shown in FIGS. **9** and **10**.

[0031] As shown in FIG. **11**, the first flat link **191** includes an end slot **194** at the second end, which pushes on the second panel **172** during expansion. The inner surface of the second panel includes a

latch **195** for securing the first flat link **191**. The latch **195** is inserted in the end slot **194** in the expanded position to stiffen the expanded body and lock the body in the expanded position.

[0032] A second set of flat links **196, 197** is similarly connected to the panels **171, 172** and each other at an outer side of the body **161**.

[0033] The illustrative linkage design minimizes the thickness of the body when the panels are in the collapsed position and facilitates selected rotation of the body **161** relative to the mover **133**. Other linkage designs for selectively expanding the body may be used.

[0034] Referring to FIG. **12**, an actuator **182** for the expandable rod **180** may be housed in the sheath **164**. The actuator **182** may be powered from pickup coils **136** mounted in the mover **133** along with necessary rectifiers and control electronics **137**, which connect to the actuator **182**. The pickup coils derive power either from the same coils that drive the movers **133** or independent coils mounted alongside the mover coils. Actuator commands could also pass over the pickup coils or could be delivered wirelessly.

[0035] The invention has been described relative to certain illustrative embodiments. Those skilled in the art will appreciate that the present invention may be implemented in a number of different applications and embodiments and is not specifically limited in its application to the particular embodiments depicted.

Claims

1. A flight for a conveyor comprising: an expandable body comprising a first panel, a second panel and a linkage system between the first panel and the second panel for selectively moving the second panel relative to the first panel; and a mounting portion connected to the first panel for mounting the expandable body relative to the conveyor.
2. The flight of claim 1, wherein the mounting portion comprises a mounting bar rotatably mounted in a sheath.
3. The flight of claim 2, wherein the sheath is mounted on a mover of a linear magnetic motor.
4. The flight of claim 2, wherein the mounting bar houses a retractable rod connected to the linkage system and an actuator for selectively extending and retracting the retractable rod.
5. The flight of claim 4, wherein the actuator is powered from pickup coils mounted in a mover of a linear magnetic motor to which the mounting portion is mounted.
6. The flight of claim 1, wherein the linkage system comprises: a first link pivotally connected at a first end to a retractable rod, the first link pivotally connected to a lateral middle portion of the second panel at a second end; and a second link pivotally connected at a first end to a middle portion of the first link and at a second end to a middle portion of the first panel.
7. The flight of claim 6, wherein the retractable rod is housed in a sheath fixed to the first panel.
8. The flight of claim 6, wherein the linkage system further comprises a set of hingedly connected flat links below the first link and second link.
9. The flight of claim 8, wherein a first flat link includes an opening for latching onto a tab on an inner surface of the second panel.
10. The flight of claim 1, wherein the first panel includes a follower roller extending from a distal end for engaging a track on the conveyor to selectively rotate the flight.
11. A conveyor system for conveying and spacing articles, comprising: a conveyor having a conveying surface extending longitudinally from a first end to a second end; and a plurality of flights extending laterally over the conveying surface, each flight having an expandable body that can expand longitudinally.
12. The conveyor system of claim 11, wherein each flight comprises: an expandable body comprising a first panel, a second panel and a linkage system between the first panel and the second panel for selectively moving the second panel relative to the first panel; and a mounting portion connected to the first panel for mounting the expandable body relative to the conveyor.

13. The conveyor system of claim 12, wherein each mounting portion comprises a mounting bar rotatably mounted in a sheath.

14. The conveyor system of claim 13, wherein the sheath is mounted on a mover of a linear magnetic motor.

15. The conveyor system of claim 11, wherein the conveyor includes a frame having a track and each flight includes a follower roller for engaging the track to control the rotational position of the flight.
